

Bangladesh University of Business & Technology (BUBT)



LAB Report

Course Code : CSE 210
Course Title : Operating Systems Lab
Experiment No. : 04

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Intake: 53
Section: 01
Program: B.Sc. Engg. in CSE

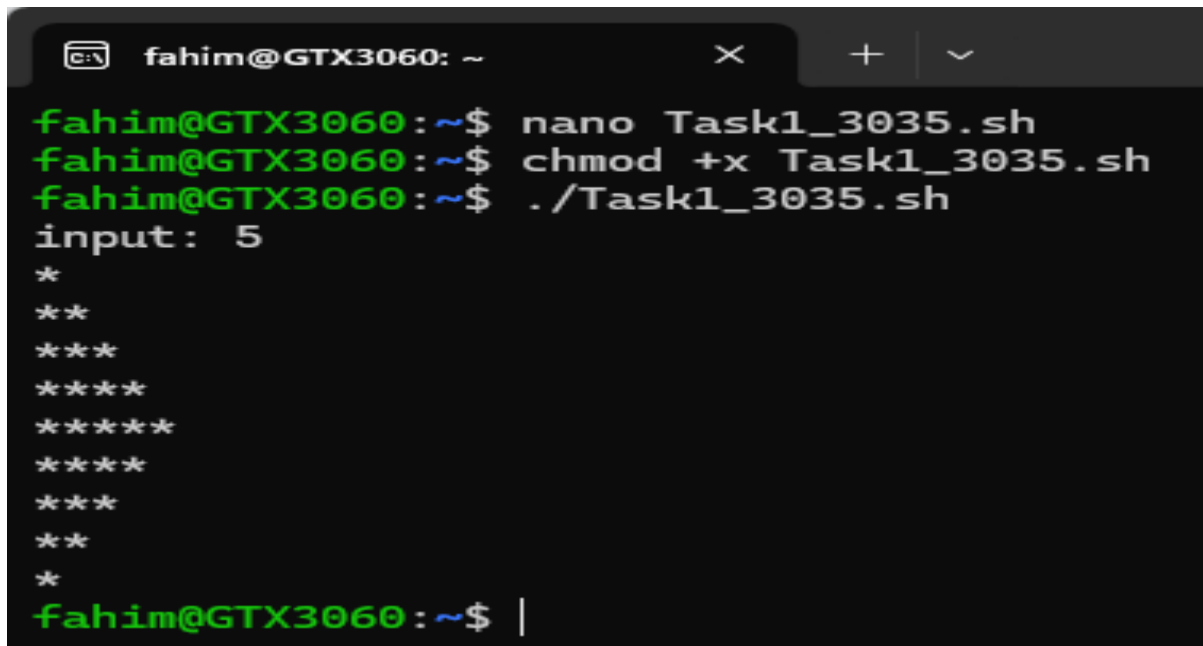
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Date of Submission: 24/08/2025

Signature of Teacher

1. Print a bash script to generate the following pattern.

```
nano Task1_3035.sh
chmod +x Task1_3035.sh
./Task1_3035.sh
```

A terminal window titled 'fahim@GTX3060: ~' with standard window controls. The user enters three commands: 'nano Task1_3035.sh', 'chmod +x Task1_3035.sh', and './Task1_3035.sh'. The script prompts for 'input: 5'. It then prints a pattern of asterisks: a single line of 5 asterisks, followed by lines of 4, 3, 2, and 1 asterisk, and finally a single line of 5 asterisks. The prompt 'fahim@GTX3060: ~\$ |' is visible at the bottom.

```
fahim@GTX3060: ~$ nano Task1_3035.sh
fahim@GTX3060: ~$ chmod +x Task1_3035.sh
fahim@GTX3060: ~$ ./Task1_3035.sh
input: 5
*
**
***
****
*****
*****
****
***
**
*
fahim@GTX3060: ~$ |
```

Inside 20245103035_Info.sh:

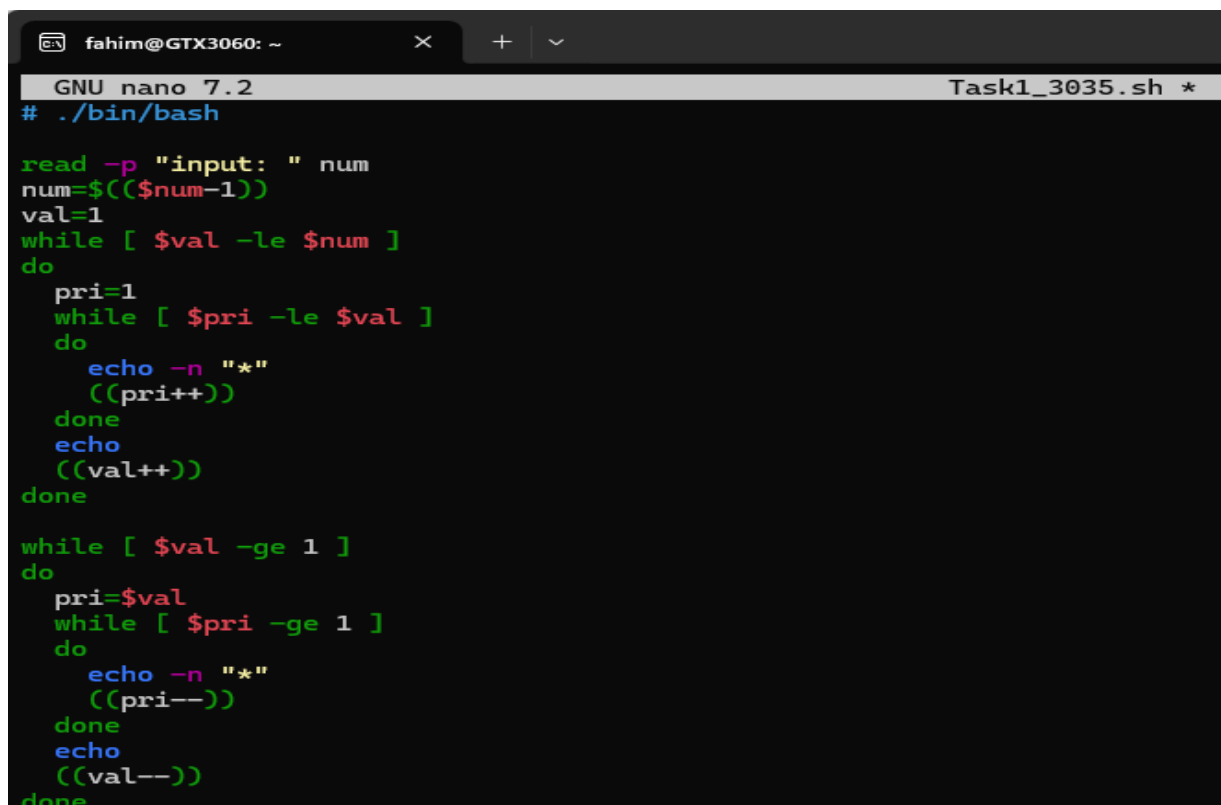
```
# ./bin/bash
read -p "input: " num
num=$((num-1))
val=1
while [ $val -le $num ]
do
    pri=1
    while [ $pri -le $val ]
    do
        echo -n "*"
        ((pri++))
    done
    echo
    ((val++))
done

while [ $val -ge 1 ]
do
    pri=$val
```

```

while [ $pri -ge 1 ]
do
    echo -n "*"
    ((pri--))
done
echo
((val--))
done

```



```

fahim@GTX3060: ~
GNU nano 7.2 Task1_3035.sh *
# ./bin/bash

read -p "input: " num
num=$((num-1))
val=1
while [ $val -le $num ]
do
    pri=1
    while [ $pri -le $val ]
    do
        echo -n "*"
        ((pri++))
    done
    echo
    ((val++))
done

while [ $val -ge 1 ]
do
    pri=$val
    while [ $pri -ge 1 ]
    do
        echo -n "*"
        ((pri--))
    done
    echo
    ((val--))
done

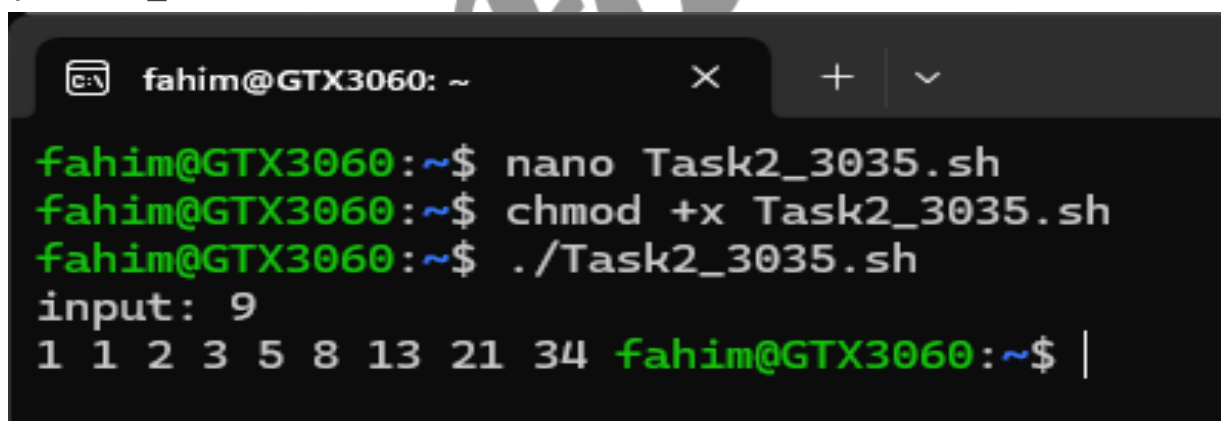
```

2. Print the Fibonacci series in terminal with a bash script.

```

nano Task2_3035.sh
chmod +x Task2_3035.sh
./Task2_3035.sh

```



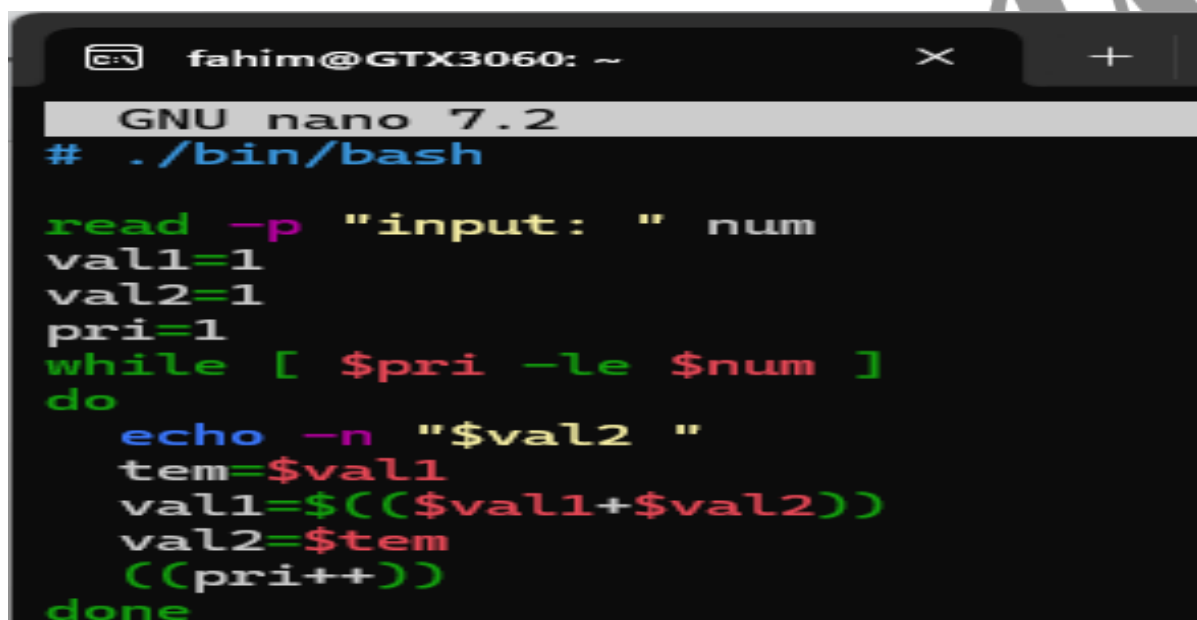
```

fahim@GTX3060: ~$ nano Task2_3035.sh
fahim@GTX3060: ~$ chmod +x Task2_3035.sh
fahim@GTX3060: ~$ ./Task2_3035.sh
input: 9
1 1 2 3 5 8 13 21 34 fahim@GTX3060: ~$ |

```

Inside 20245103035_Operations.sh:

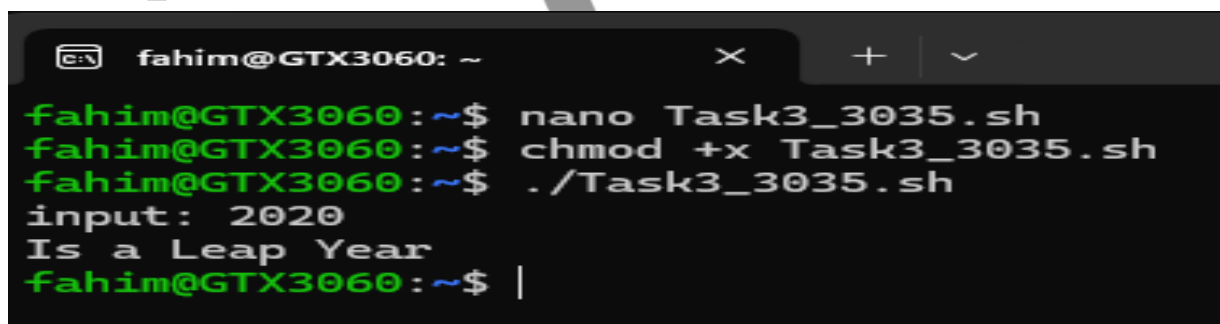
```
# ./bin/bash
read -p "input: " num
val1=1
val2=1
pri=1
while [ $pri -le $num ]
do
    echo -n "$val2 "
    tem=$val1
    val1=$(( $val1+$val2 ))
    val2=$tem
    ((pri++))
done
```

A screenshot of a terminal window titled 'fahim@GTX3060: ~'. The window shows the GNU nano 7.2 editor with the following bash script:

```
# ./bin/bash
read -p "input: " num
val1=1
val2=1
pri=1
while [ $pri -le $num ]
do
    echo -n "$val2 "
    tem=$val1
    val1=$(( $val1+$val2 ))
    val2=$tem
    ((pri++))
done
```

3. Write a bash script to find if a year is a leap year or not.

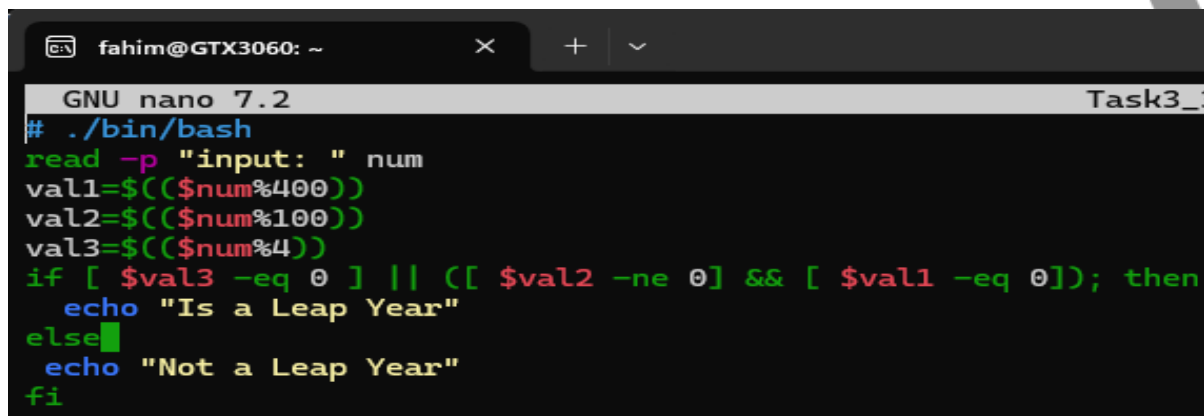
```
nano Task3_3035.sh
chmod +x Task3_3035.sh
./Task3_3035.sh
```

A screenshot of a terminal window titled 'fahim@GTX3060: ~'. It shows the execution of the Task3_3035.sh script:

```
fahim@GTX3060:~$ nano Task3_3035.sh
fahim@GTX3060:~$ chmod +x Task3_3035.sh
fahim@GTX3060:~$ ./Task3_3035.sh
input: 2020
Is a Leap Year
fahim@GTX3060:~$ |
```

Inside 20245103035_AgeCheck.sh:

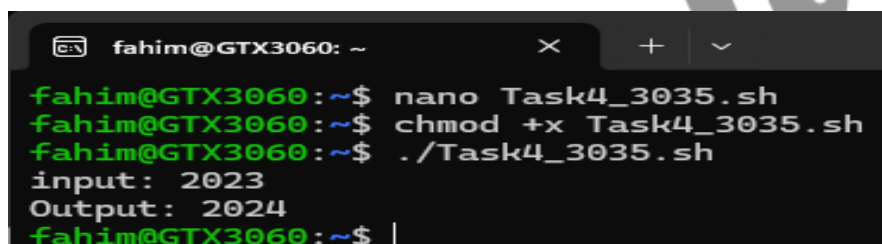
```
# ./bin/bash
read -p "input: " num
val1=$(( $num % 400 ))
val2=$(( $num % 100 ))
val3=$(( $num % 4 ))
if [ $val3 -eq 0 ] || ([ $val2 -ne 0 ] && [ $val1 -eq 0 ]); then
    echo "Is a Leap Year"
else
    echo "Not a Leap Year"
fi
```

A screenshot of a terminal window with a dark background. The window title is 'fahim@GTX3060: ~'. The terminal shows the GNU nano 7.2 editor with the following script content:

```
# ./bin/bash
read -p "input: " num
val1=$(( $num % 400 ))
val2=$(( $num % 100 ))
val3=$(( $num % 4 ))
if [ $val3 -eq 0 ] || ([ $val2 -ne 0 ] && [ $val1 -eq 0 ]); then
    echo "Is a Leap Year"
else
    echo "Not a Leap Year"
fi
```

4. Write a bash script to find the nearest leap year from the current year.

```
nano Task4_3035.sh
chmod +x Task4_3035.sh
./Task4_3035.sh
```

A screenshot of a terminal window with a dark background. The window title is 'fahim@GTX3060: ~'. The terminal shows the following commands and output:

```
fahim@GTX3060:~$ nano Task4_3035.sh
fahim@GTX3060:~$ chmod +x Task4_3035.sh
fahim@GTX3060:~$ ./Task4_3035.sh
input: 2023
Output: 2024
fahim@GTX3060:~$ |
```

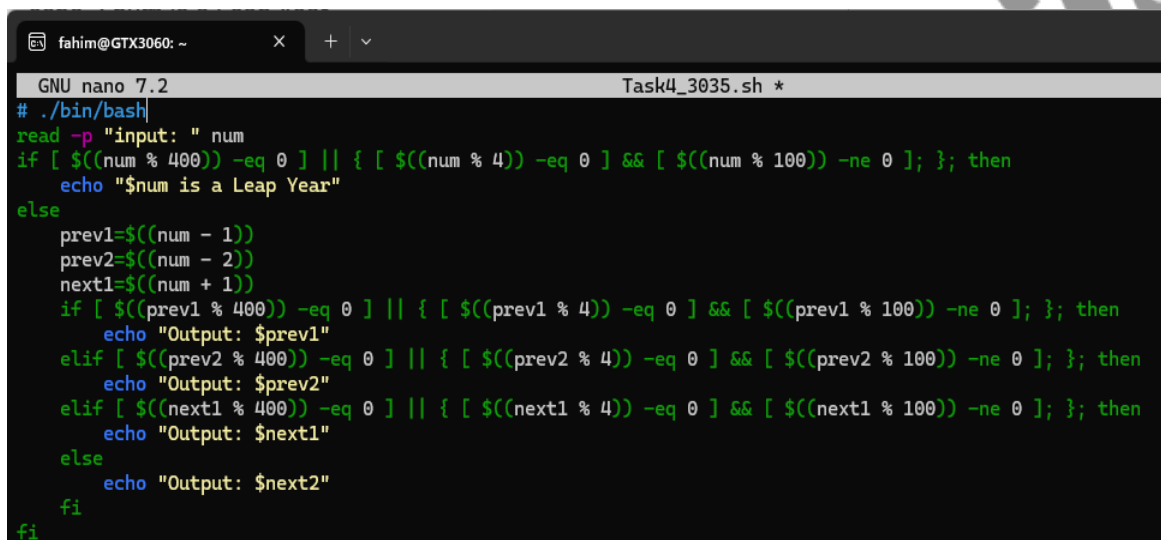
Inside 20245103035_ArrayAccess.sh:

```
# ./bin/bash
read -p "input: " num
if [ $((num % 400)) -eq 0 ] || { [ $((num % 4)) -eq 0 ] && [ $((num % 100)) -ne 0 ]; }; then
    echo "$num is a Leap Year"
else
```

```

prev1=$((num - 1))
prev2=$((num - 2))
next1=$((num + 1))
if [ $((prev1 % 400)) -eq 0 ] || { [ $((prev1 % 4)) -eq 0 ] && [
    $((prev1 % 100)) -ne 0 ]; }; then
    echo "Output: $prev1"
elif [ $((prev2 % 400)) -eq 0 ] || { [ $((prev2 % 4)) -eq 0 ] && [
    $((prev2 % 100)) -ne 0 ]; }; then
    echo "Output: $prev2"
elif [ $((next1 % 400)) -eq 0 ] || { [ $((next1 % 4)) -eq 0 ] && [
    $((next1 % 100)) -ne 0 ]; }; then
    echo "Output: $next1"
else
    echo "Output: $next2"
fi
fi

```



```

GNU nano 7.2 Task4_3035.sh *
# ./bin/bash
read -p "input: " num
if [ $((num % 400)) -eq 0 ] || { [ $((num % 4)) -eq 0 ] && [ $((num % 100)) -ne 0 ]; }; then
    echo "$num is a Leap Year"
else
    prev1=$((num - 1))
    prev2=$((num - 2))
    next1=$((num + 1))
    if [ $((prev1 % 400)) -eq 0 ] || { [ $((prev1 % 4)) -eq 0 ] && [ $((prev1 % 100)) -ne 0 ]; }; then
        echo "Output: $prev1"
    elif [ $((prev2 % 400)) -eq 0 ] || { [ $((prev2 % 4)) -eq 0 ] && [ $((prev2 % 100)) -ne 0 ]; }; then
        echo "Output: $prev2"
    elif [ $((next1 % 400)) -eq 0 ] || { [ $((next1 % 4)) -eq 0 ] && [ $((next1 % 100)) -ne 0 ]; }; then
        echo "Output: $next1"
    else
        echo "Output: $next2"
    fi
fi

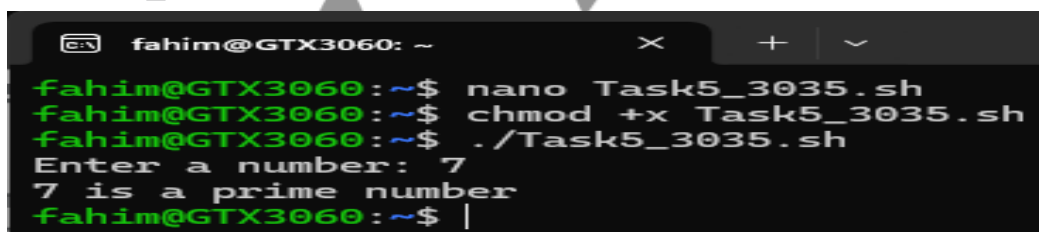
```

5. Write a bash script to check whether a given number is a prime number..

```

nano Task5_3035.sh
chmod +x Task5_3035.sh
./Task5_3035.sh

```



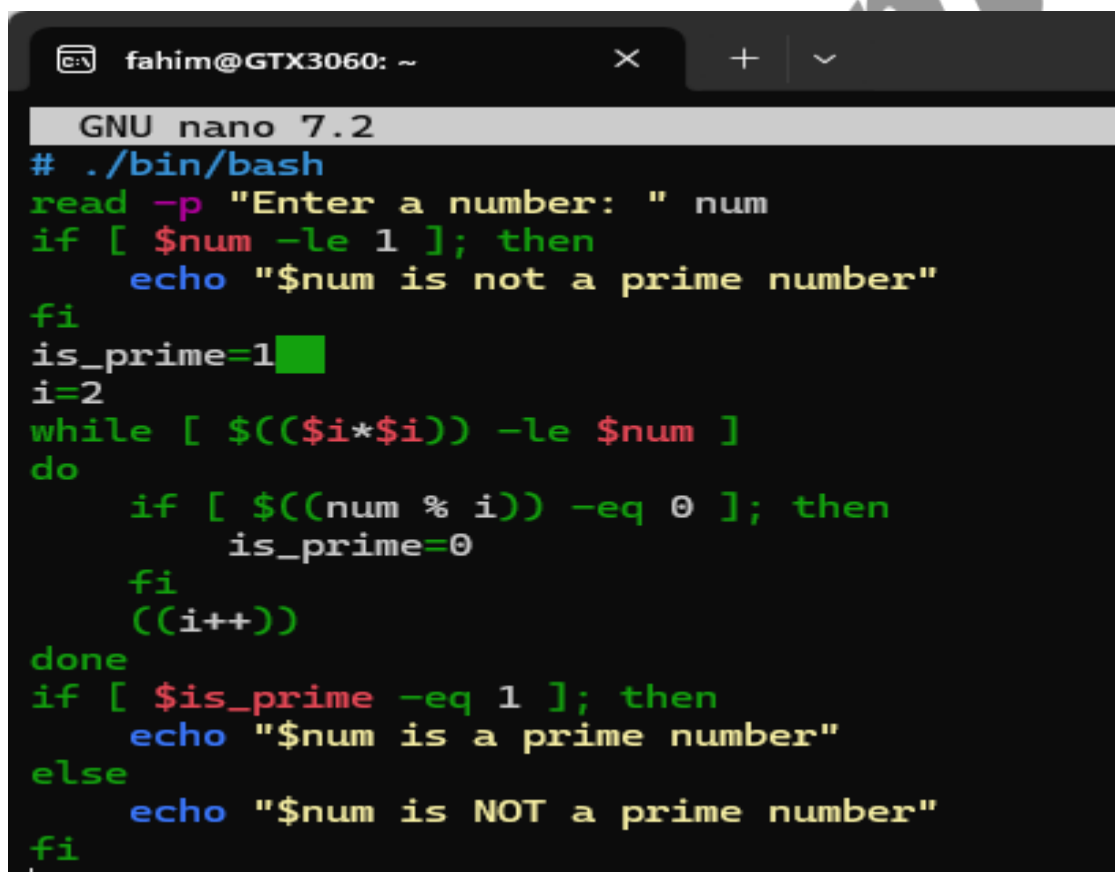
```

fahim@GTX3060: ~$ nano Task5_3035.sh
fahim@GTX3060: ~$ chmod +x Task5_3035.sh
fahim@GTX3060: ~$ ./Task5_3035.sh
Enter a number: 7
7 is a prime number
fahim@GTX3060: ~$ |

```

Inside 20245103035_ArrayAccess.sh:

```
# ./bin/bash
read -p "Enter a number: " num
if [ $num -le 1 ]; then
    echo "$num is not a prime number"
fi
is_prime=1
i=2
while [ $((i*i)) -le $num ]
do
    if [ $(num % i) -eq 0 ]; then
        is_prime=0
    fi
    ((i++))
done
if [ $is_prime -eq 1 ]; then
    echo "$num is a prime number"
else
    echo "$num is NOT a prime number"
fi
```

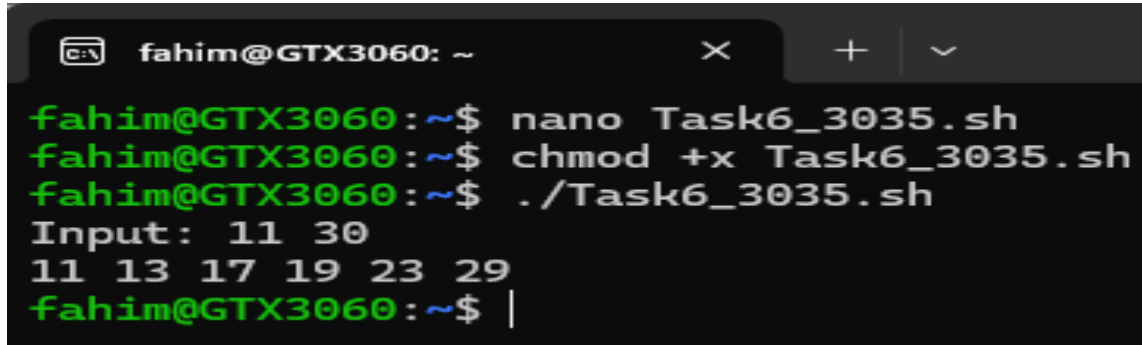


The screenshot shows a terminal window with the title bar 'fahim@GTX3060: ~'. The terminal is running GNU nano 7.2. The script content is displayed in a monospaced font with syntax highlighting: blue for comments, green for keywords, red for variables, and yellow for strings. A cursor is visible at the end of the line 'is_prime=1'.

```
GNU nano 7.2
# ./bin/bash
read -p "Enter a number: " num
if [ $num -le 1 ]; then
    echo "$num is not a prime number"
fi
is_prime=1
i=2
while [ $((i*i)) -le $num ]
do
    if [ $(num % i) -eq 0 ]; then
        is_prime=0
    fi
    ((i++))
done
if [ $is_prime -eq 1 ]; then
    echo "$num is a prime number"
else
    echo "$num is NOT a prime number"
fi
```

6. Write a bash script to find and print all prime numbers between two given numbers (inclusive)..

```
nano Task6_3035.sh
chmod +x Task6_3035.sh
./Task6_3035.sh
```

A terminal window with a dark background and light green text. The prompt is 'fahim@GTX3060: ~'. The user enters 'nano Task6_3035.sh', then 'chmod +x Task6_3035.sh', and finally './Task6_3035.sh'. The script prompts for input: 'Input: 11 30'. It then outputs the prime numbers: '11 13 17 19 23 29'. The prompt returns to 'fahim@GTX3060: ~\$ |'.

Inside 20245103035_ArrayAccess.sh:

```
# ./bin/bash
read -p "Input: " start end
num=$start
while [ $num -le $end ]
do
if [ $num -le 1 ]; then
    ((num++))
else
is_prime=1
i=2
while [ $((i*i)) -le $num ]
do
    if [ $((num % i)) -eq 0 ]; then
        is_prime=0
    fi
    ((i++))
done

if [ $is_prime -eq 1 ]; then
    echo -n "$num "
fi
fi
((num++))
```


done
echo

```
fahim@GTX3060: ~
GNU nano 7.2
# ./bin/bash
read -p "Input: " start end
num=$start
while [ $num -le $end ]
do
if [ $num -le 1 ]; then
    ((num++))
else
is_prime=1
i=2
while [ $$i*$i) -le $num ]
do
    if [ $(num % i) -eq 0 ]; then
        is_prime=0
    fi
    ((i++))
done
if [ $is_prime -eq 1 ]; then
    echo -n "$num "
fi
fi
((num++))
done
echo
```